Day 1 — Introduction to Full Stack Development

Simple Questions, Answers, and Client–Server Architecture

1. What is Full Stack Development?

Full Stack Development means building both the frontend and backend of a web application. It also involves working with a database to store and manage information. A Full Stack Developer can work on the complete application, from the user interface to the server and database.

2. What is the difference between Frontend, Backend, and Database?

Part	Purpose	Examples
Frontend	The part users see and interact with.	HTML, CSS, JavaScript, React.js
Backend	Processes requests, applies business rules, and sends responses.	Java, Spring Boot
Database	Stores and retrieves application data.	SQL, MySQL, PostgreSQL

3. What is Client–Server Architecture?

Client–server architecture is a model in which a client (such as a web browser) sends a request to a server. The server processes the request and may read or update information in a database. It then sends a response back to the client.

Typical flow: User action → request → server logic → database → response

4. How does a web application work?

1. The user opens a website in a browser and performs an action, such as clicking Login.
2. The frontend sends a request to the backend server.
3. The backend checks the request and runs the required logic.
4. If needed, the backend reads or updates data in the database.
5. The backend sends a response, and the frontend displays the result to the user.

5. What is Java Full Stack Development?

Java Full Stack Development means creating a complete web application using frontend technologies and Java-based backend technologies. For example, the frontend may use HTML, CSS, JavaScript, and React.js; the backend may use Java and Spring Boot; and the database may use SQL with MySQL.

6. What are the main technologies used?

Technology	Use
HTML	Defines the structure and content of a webpage.
CSS	Controls the layout, colours, spacing, and appearance.
JavaScript	Adds interaction and dynamic behaviour to webpages.
React.js	Builds reusable frontend components and user interfaces.
Java	A programming language commonly used for backend development.
Spring Boot	Helps build Java backend applications and REST APIs.
SQL	A language used to query and manage relational databases.
MySQL	A relational database management system that stores application data.

7. Example of a Java Full Stack Application

Imagine a student registration website. React.js displays the registration form. When the student submits it, the frontend sends the information to a Spring Boot API. The backend validates the data and stores it in a MySQL database using SQL. The backend returns a success or error message, which the frontend shows to the student.

8. Quick Revision

• **Full Stack Development:** Working with frontend, backend, and database.

• **Frontend:** What the user sees and interacts with.

• **Backend:** Handles application logic and requests.

• **Database:** Stores and retrieves data.

• **Java Full Stack example:** HTML + CSS + JavaScript/React.js + Java/Spring Boot + SQL/MySQL.